

MetaPhlAn-Microbiome-Analysis

Sample File

Downloaded in batches. Example:

rsync -aP /Users/milana/Desktop/drive-download-20260417T061612Z-3-001.zip
mserkin@hpc-student.charlotte.edu:/users/mserkin/lab8_metaphlan/

Running Metaphlan2 on all files at the same time:

```
GNU nano 5.6.1 run_metaphlan.sh
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=16
#SBATCH --mem=64G
#SBATCH --time=12:00:00

# Load MetaPhlan
module load metaphlan/4.2.2

# 1. Merge the paired-end files (Commented out because it is already done!)
# for file1 in ./R1*; do
#     file2=${file1}/R2
#     zcat ${file1} ${file2} > ${file1%R1*}merged.fastq
# done

# 2. Run MetaPhlan on all merged files (using 16 cores)
for f in *merged.fastq; do
    # NOTE: Changed multifastq to fastq here!
    metaphlan $f --input_type fastq --nproc 16 > ${f%merged.fastq}_profile.txt
done

# 3. Merge all the individual results into one master table
merge_metaphlan_tables.py *_profile.txt > merged_abundance_table.txt

# 4. Clean up the table to only show the Species-level data
grep -E "(s__)|(^ID)" merged_abundance_table.txt | grep -v "t__" | sed 's/^.*s__//g' > merged_abundance_table_species.txt
```

Making a Heatmap:

Heatmaps are a typical way of visualizing taxonomic clusters to get an idea about presence and absence of species across multiple samples. The heatmap gives you an idea of which species are abundant in each location, and can also give a clue to when samples in a group of replicates are outliers. In this set of samples, the third UPA replicate does not cluster with the other two, which suggests that there may be a problem with that replicate.

Questions

1. Based on the heatmap you created, which samples are most similar to each other in their microbial community? On what basis can you determine this from the heatmap?

Based on the heatmap, I can determine which samples are most similar to each other by looking at the branching tree structures(dendrogram) at the top of the heatmap. The samples that are connected by the shortest, lowest branches are the most similar according to the Bray Curtis distances. According to this, we can see that each group of replicates (ATE, DSA, UPA, and RES) are most similar to one another. There is a pattern we can observe that out of each group of replicates, has 2 clusters: (UPA_64D2,UPA_65D2) and (UPA_66D2), (RES_38, RES_39) and (RES_37), (ATE_49) and (ATE_50, ATE_51), (DSA_58) and (DSA_59, DSA_60). The () represent clusters. Based on the branching, we can also conclude that ATE and DSA have environments that are similar in microbial community structures.

2. Compare the combined abundance tables for summarization at the phylum level, and at the genus and species level. What do you notice about this data? How common are zero values at each level of summarization? How common are very low abundances?

Description: df [3 × 4]

Level <chr>	Total_Taxa <int>	Percent_Zero <chr>	Percent_Very_Low <chr>
Phylum	13	51.9%	9%
Genus	288	73.5%	14.7%
Species	424	74%	14.6%

I noticed that as specificity increases(phylum>genus>species), the total number of taxa, the percentage of zero values and the low abundances all increase. This tells us that even though the identified taxa increase, the data becomes more sparse. A reason for this pattern could be that the specific bacterial species are highly specialized only in their local environments and arent found elsewhere. Low abundance increasing tells us that the environments have a small amount of highly abundant species.

3. Another way to visualize the similarities and differences between groups of samples is a PCoA. Does this plot tell you anything that you can not already see from the heatmap?

I used R to make a PCoA plot. The 95.2% of the total variance (PCoA 1: 52.4% + PCoA 2: 42.8%) confirms that the location the species are found is the main reason for the microbiome. It also confirms the findings in the heatmap that DSA and ATE are more similar to each other as they are closer together than the rest of the groups. Looking at PCoA1, it also looks like RES is the furthest away group around 0.56 PCoA1, UPA is around 0.15 PCoA1, and ATE and DSA hover around -0.35 PCoA1. Based on PCoA2, it also looks like UPA is the furthest away group

as it is above 0.5 PCoA2 while the other 3 groups are below 0 PCoA2. This plot also reveals that ATE and DSA have more variance on PCoA2 than PCoA1.

4. Include your plots and your R code for producing a PCoA

